Run report: join-q7-run-20261024T2258Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part JOIN under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: join-q7 20261024T2258Z
git HEAD: b08305739b2f95bb83966c2830da5e8d14eb2d92
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 26415a19d8d07de211c7992a6c97ca99a2a3a358fabb5c02e7893639121efd5b
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/join-q7-run-20261024T2258Z JOIN 7 research/genus4_n4_census_20261024/results/cen-q7-cone-run-20261024T2241Z/
mass_by_t.csv,research/genus4_n4_census_20261024/results/cen-q7-nonsplit-run-20261024T2243Z/mass_by_t.csv,research/genus4_n4_census_20261024/results/cen-q7-split-run-20261024T2239Z/mass_by_t.csv
research/genus4_n4_census_20261024/results/tian-q7-run-20261024T2239Z/tian_by_t.csv
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 0
```

tool_log.txt

```
q=7: prime-order traces with an elliptic curve: 20 (ordinary 20); traces with no hit: []
Spearman(mass, H) = 0.386
Spearman(mass, Haar density) = 0.636
Spearman(mass, Haar*H/sqrt(disc)) = 0.579
Spearman(mass, #l with Tian kernel (depends = 1/2)) = 0.217
fit mass = c*H: c = 4.295237e0, R^2 (through origin, ratio-of-sums c) = -0.023
fit mass = c*haar: c = 1.233694e3, R^2 (through origin, ratio-of-sums c) = 0.556
fit mass = c*haar*H/sqrt: c = 5.690572e3, R^2 (through origin, ratio-of-sums c) = 0.338
```

join_summary.txt

```
q=7: prime-order traces with an elliptic curve: 20 (ordinary 20); traces with no hit: []
Spearman(mass, H) = 0.386
Spearman(mass, Haar density) = 0.636
Spearman(mass, Haar*H/sqrt(disc)) = 0.579
Spearman(mass, #l with Tian kernel (depends = 1/2)) = 0.217
fit mass = c*H: c = 4.295237e0, R^2 (through origin, ratio-of-sums c) = -0.023
fit mass = c*haar: c = 1.233694e3, R^2 (through origin, ratio-of-sums c) = 0.556
fit mass = c*haar*H/sqrt: c = 5.690572e3, R^2 (through origin, ratio-of-sums c) = 0.338
```

join_by_t.csv (first 12 of 20 rows)

t	nE	ordinary	hits	mass	mass_se	H	haar_density	arith_x_haar	tian_yes	tian_depends	tian_no
−75	2477	true	164	93.4391	8.1300	20.000	0.048655	0.015427	2	0	1
−71	2473	true	143	90.0071	8.1374	17.333	0.055147	0.014151	3	0	0
−65	2467	true	111	77.4495	7.7089	24.000	0.063845	0.020892	2	0	1
−57	2459	true	159	88.3692	7.8636	16.000	0.073555	0.014763	2	0	1
−45	2447	true	230	134.6486	9.8148	16.000	0.084258	0.015485	2	0	1
−39	2441	true	108	70.4493	7.2406	16.000	0.087955	0.015653	2	0	1
−15	2417	true	165	88.4667	7.7950	24.000	0.092688	0.022970	2	0	1
−9	2411	true	123	76.8918	7.5317	12.000	0.091579	0.011261	2	0	1
3	2399	true	228	137.8221	10.0039	16.000	0.086943	0.014201	2	0	1
9	2393	true	181	104.7882	8.6610	12.000	0.083522	0.010271	3	0	0
13	2389	true	191	108.6797	8.7819	32.000	0.080869	0.026642	2	0	1
19	2383	true	264	152.6362	10.4356	20.000	0.076376	0.015888	3	0	0

Graph

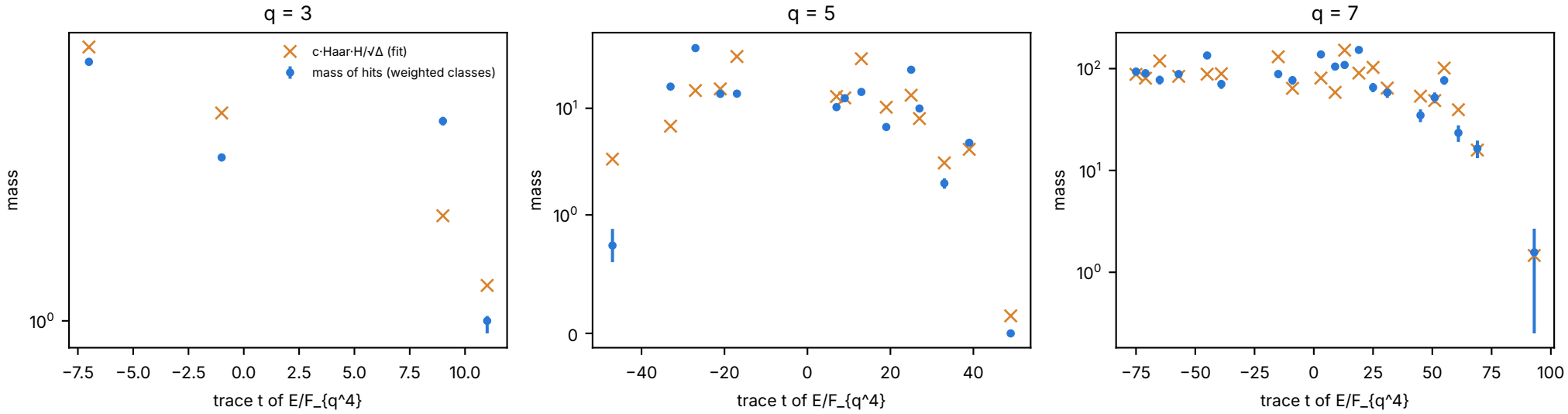


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.